From Selection to Scheduling: Federated Geometry-Aware Correction Makes Exemplar Replay Work Better under Continual Dynamic Heterogeneity

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Abstract

Exemplar replay has become an effective strategy for mitigating catastrophic forgetting in federated continual learning (FCL) by retaining representative samples from past tasks. Existing studies focus on designing sample-importance estimation mechanisms to identify information-rich samples. However, they typically overlook strategies for effectively utilizing the selected exemplars, which limits their performance under continual dynamic heterogeneity across clients and tasks. To address this issue, this paper proposes a Federated gEometry-Aware correcTion method, termed FEAT*, which alleviates imbalance-induced representation collapse that drags rare-class features toward frequent classes across clients. Specifically, it consists of two key modules: 1) the Geometric Structure Alignment module performs structural knowledge distillation by aligning the pairwise angular similarities between feature representations and their corresponding Equiangular Tight Frame prototypes, which are fixed and shared across clients to serve as a class-discriminative reference structure. This encourages geometric consistency across tasks and helps mitigate representation drift; 2) the Energy-based Geometric Correction module removes task-irrelevant directional components from feature embeddings, which reduces prediction bias toward majority classes. This improves sensitivity to minority classes and enhances the model's robustness under class-imbalanced data distributions. Experimental results demonstrate that* FEAT *mitigates dual bias at both the client level and the task level, outperforming seven state-of-the-art methods.*

1. Introduction

Federated learning (FL) is a distributed machine learning framework that enables multiple clients to collaboratively train a shared model without sharing their raw data [11, 13, 27, 34]. It enables periodic model parameter exchanges between clients and the server, where the server aggregates local models from clients to iteratively optimize the global model [3, 8, 12, 44]. In real-world FL applications, edge devices are required to handle a continuously evolving sequence of tasks, such as the continual arrival of novel classes, rather than dealing with a static single-task setting [1, 46, 49]. This non-stationary data environment poses a challenge for models to achieve continual learning. Fundamentally, these challenges lie in two aspects: 1) limited local storage hinders the retention of past knowledge; 2) the inherent data heterogeneity across clients, along with the dynamically evolving data distributions, degrades the performance of collaborative learning. These factors lead to severe catastrophic forgetting in federated continual learning (FCL) [40, 48].

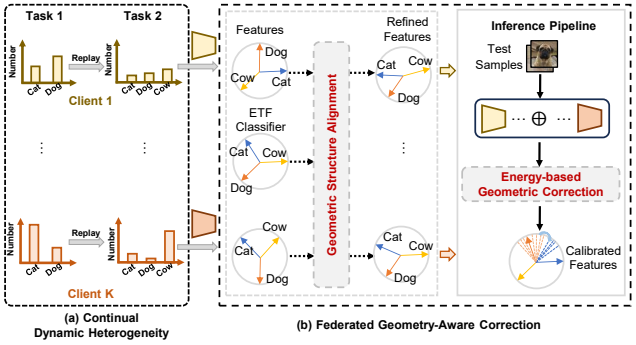

Figure 1. (a) Continual dynamic heterogeneity: clients see different class distributions over time, and each new task is learned with limited replay of past data. (b) Federated geometry-aware calibration: FEAT aligns client feature geometry and applies an energy-based correction at inference to calibrate prediction.

To mitigate catastrophic forgetting, existing studies have widely adopted replay-based methods due to their simplicity and effectiveness. Based on the implementation of replay mechanisms, existing methods can be broadly categorized into two types. (1) Generative-based replay methods utilize generative models, such as Generative Adversarial Networks [4, 28], to synthesize pseudo-samples that resemble historical data [33, 37, 45]. These methods train a gen-

erator to approximate the distribution of previously encountered data, which can enhance privacy and reduce memory usage [43, 50]. While conceptually appealing, they typically face significant challenges, including the high computational cost associated with training generative models and the potential suboptimal quality of generated samples [43, 45]. (2) In contrast, exemplar-based replay methods offer inherent advantages in terms of knowledge fidelity, as they avoid the challenges associated with training generative models. They typically store a small portion of the data from previous tasks in local memory and replay it during the training of new tasks [6, 19]. Despite the performance advantages demonstrated in existing studies, most of them primarily focus on how to select representative samples, while paying less attention to how to efficiently utilize these limited exemplars under resource constraints [20].

To address these issues, this paper proposes a federated geometry-aware correction method for class-incremental learning, termed FEAT, which mitigates the optimization bias arising from inter-client data heterogeneity and from the imbalance between majority and minority classes. Figure 1 illustrates the main idea of FEAT. Specifically, it is designed with two essential components: (1) the geometric structure alignment module enforces angular consistency between local feature representations and globally shared prototypes by distilling relational geometry in the embedding space. By aligning intra-batch feature correlations with those of fixed Equiangular Tight Frame prototypes, it regularizes the local learning dynamics and fosters structurally consistent representations across clients, which can enhance generalization under heterogeneous task distributions. (2) Moreover, to alleviate task-level data imbalance, the energy-based geometric correction module removes task-irrelevant components from the feature space during inference, which reduces overconfidence in majority classes and enhancing the model's sensitivity to underrepresented classes. By jointly leveraging geometric distillation and debiasing, FEAT effectively harmonizes local and global objectives, yielding improved performance under heterogeneous and imbalanced cases.

Extensive experiments were conducted on three datasets with varying levels of heterogeneity, including performance comparison, ablation study, sensitivity analysis of hyperparameters, and case study on the working mechanism of key modules. The results validate that FEAT effectively enhances the consistency of representation learning across clients and improves model robustness under long-tailed data distributions. Across all benchmarks, FEAT outperforms seven state-of-the-art methods, showing consistent gains in Top-1 accuracy. In summary, this paper makes the following key contributions:

- This paper reveals two key challenges that persist in exemplar replay-based FCL: the use of replayed data exacerbates inter-client heterogeneity and leads to distributional imbalance between past and current task data.
- This study proposes a method orthogonal to exemplar-replay policies, enabling seamless composition with different strategies without changing selection criteria or memory allocation.
- The results validate that the prototypical angular distillation improves inter-client feature consistency, the normalized de-biasing classifier alleviates class imbalance during inference, jointly addressing these challenges.

2. Related Work

2.1. Generative Replay for FCL

Generative replay trains models such as Variational Autoencoders (VAEs [9, 32]) or Generative Adversarial Networks (GANs [10]) to approximate the sample distribution of previous tasks, so the learner can revisit pseudo-samples while training new tasks and thus mitigate catastrophic forgetting [2, 24, 29, 30, 43, 45, 47, 48, 50]. In FCL, LANDER uses label text embeddings from a pretrained language model to guide data generation [43], and GenFCIL uses a lightweight server side generator to synthesize feature representations of old classes from shared class information, reducing privacy risks and memory costs by synthesizing rather than storing data [5]. However, training reliable generators often requires substantial computation and diverse data, which may be infeasible for edge devices, and generated samples may be of limited quality, which can weaken their ability to prevent forgetting [43, 50].

2.2. Exemplar Replay for FCL

Given the limits of generative models, many studies use exemplar replay, which stores a small set of representative samples to mitigate forgetting. By keeping real data, exemplar replay avoids the compute cost and sample quality issues of generators, offering a simple and faithful way to retain past knowledge [6, 19, 36, 42, 51]. For example, Re-Fed [19] and Re-Fed+ [20] learn a personalized importance model to score local samples and keep those most relevant for reducing forgetting. FedCBDR reconstructs global features of previous tasks in a privacy-preserving way, then guides class-aware and importance-sensitive sampling for balanced replay [36]. Overall, exemplar replay is practical under limited memory [6], but most works emphasize how to select samples while overlooking how to use the chosen exemplars effectively. As a result, challenges such as client heterogeneity and task-level imbalance remain [19].

2.3. Equiangular Tight Frame in FL

Equiangular Tight Frames (ETF) emerge from Neural Collapse theory, where class means form a simplex ETF and classifier weights align with these means, yielding symmet-

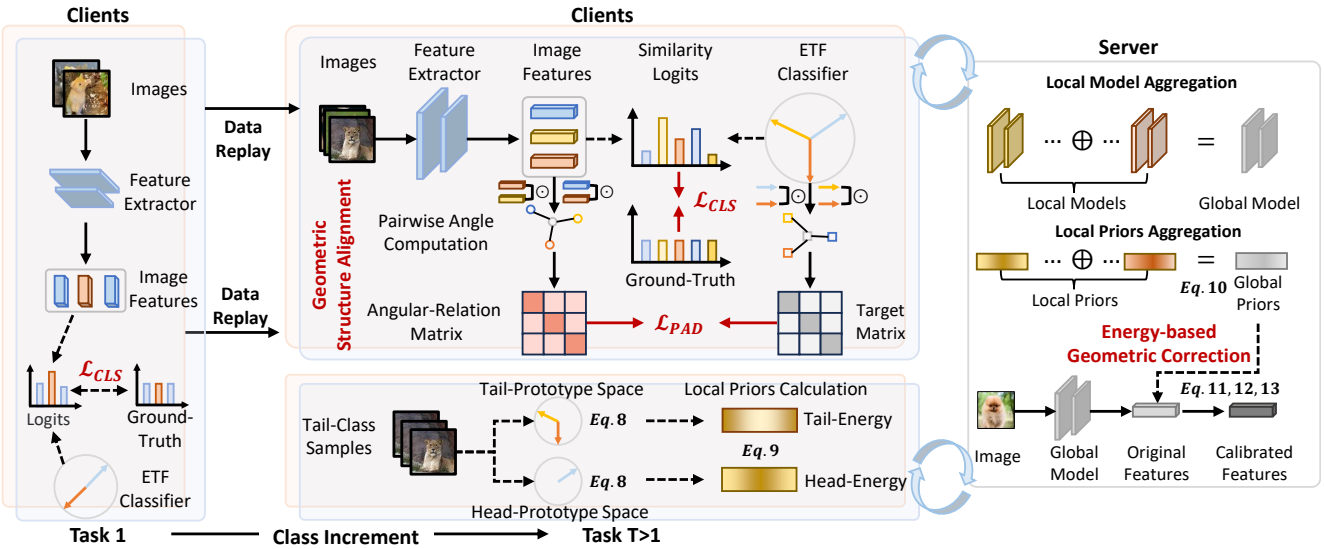

Figure 2. Illustration of the proposed FEAT framework. It contains two main modules: 1) the Geometry Structure Alignment module aligns local representations with global ETF prototypes to alleviate inter-client heterogeneity; and 2) the Energy-based Geometric Correction module removes biased components from the feature space during inference. Notably, it can be integrated into various data replay methods, including Re-Fed+ [20] and FedCBDR [36].

ric margins and stable optimization [26, 31]. In FL, several works fix a global simplex-ETF classifier to lessen classifier bias and partially align client representations under non-IID data [7, 18, 23]. Despite these advances, ETF-based approaches still face feature bias under continual dynamic heterogeneity, as shown in Figures 4 and 5.

3. Preliminaries

In federated continual learning (FCL), a central server collaborates with K distributed clients to progressively train a global model on a sequence of disjoint classification tasks. Each client k has its own task stream $\{\mathcal{D}_k^{(1)}, \mathcal{D}_k^{(2)}, \ldots, \mathcal{D}_k^{(t)}\}$, where new tasks continually introduce unseen classes. To prevent forgetting, replay-based strategies allocate a fixed-size memory buffer of capacity M on each client, which stores up to N representative samples per previous task. When a new task t arrives, the client constructs a memory set $\mathcal{B}_k^{(t-1)}$ by selecting N samples from each of the earlier tasks, ensuring that the total number of stored samples does not exceed M. The local training set at round t is then formed by merging the current data with the buffered samples, i.e., $\text{train}^{(t)} = \mathcal{D}_k^{(t)} \cup \mathcal{B}_k^{(t-1)}$. Based on these locally constructed datasets, the global model θ_t is optimized by minimizing the aggregated training loss across all clients: $\min_\theta \sum_{k=1}^{K} \sum_{(x,y)\in\text{train}^{(t)}} \mathcal{L}(f_k(x;\theta), y)$.

4. Methodology

This section presents a federated geometry-aware calibration method for data replay-based FCL, which performs debiasing across both client-level data heterogeneity and task-level data imbalance, enabling more efficient utilization of limited replay samples. The framework of the proposed method FEAT is presented in Figure 2 and Algorithm 1.

4.1. Geometric Structure Alignment (GSA)

Despite replay helps retain knowledge from previous tasks, it also introduces continually evolving heterogeneity, making it difficult for clients to maintain a consistent representation space. To alleviate this issue, we leverage a unified geometric prior by adopting the Equiangular Tight Frame (ETF) classifier, which encourages globally consistent class directions across clients [23, 38]. However, under imbalanced class distributions, cross-client alignment of tail classes (previous tasks) remains clearly weaker than that of head classes (current task), as shown in Figure 3. Therefore, the GSA module aims to enforce alignment between the angular structure of the learned features and that induced by their ETF prototypes, enabling a more robust and class-balanced representation space. Specifically, at incremental task t, we adopt a ETF over the currently observed classes $\mathcal{C}_t$ with size $C_t = |\mathcal{C}_t|$. Let $W_t = [\mathbf{w}_c]_{c\in\mathcal{C}_t} \in \mathbb{R}^{d\times C_t}$ denote the ETF prototypes for these C_t classes (with $d \geq C_t$),

$$W_t = \sqrt{\frac{C_t}{C_t - 1}}\, U_t \left(I_{C_t} - \frac{1}{C_t} \mathbf{1}\mathbf{1}^\top \right), \tag{1}$$

where $U_t \in \mathbb{R}^{d\times C_t}$ satisfies $U_t^\top U_t = I_{C_t}$, I_{C_t} is the identity, and $\mathbf{1}$ is the all-ones vector. Importantly, when a new task arrives, W_t can be reinitialized solely based on the updated class count, without requiring any data. The proto-

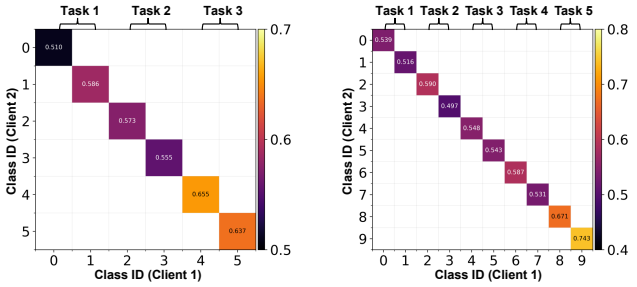

Figure 3. Cross-client feature alignment in an incremental 5-task scenario. At Task 3 (left) and Task 5 (right), cross-client alignment for earlier classes is clearly weaker than for current classes.

types have equal ℓ_2 norms and identical pair-wise angles:

$$\mathbf{w}_i^\top \mathbf{w}_j = \begin{cases} 1, & i = j, \\ -\dfrac{1}{C_t - 1}, & i \neq j, \end{cases} \qquad i, j \in \mathcal{C}_t. \tag{2}$$

Based on the above ETF prototypes, we construct cosine similarity matrices aligned with the current mini-batch. Let B denote the batch size and $\mathbf{f}_a$ the feature of the a-th sample with label $y_a \in \{1, \ldots, C_t\}$. We define:

$$M_F^{a,b} = \frac{\langle \mathbf{f}_a, \mathbf{f}_b \rangle}{\|\mathbf{f}_a\|_2 \cdot \|\mathbf{f}_b\|_2}, \; M_P^{a,b} = \frac{\langle \mathbf{w}_{y_a}, \mathbf{w}_{y_b} \rangle}{\|\mathbf{w}_{y_a}\|_2 \cdot \|\mathbf{w}_{y_b}\|_2}, \tag{3}$$

where $\mathbf{w}_{y_a}$ is the ETF prototype of the class corresponding to $\mathbf{f}_a$. Thus, both M_F and M_P are $B \times B$ matrices with the same row/column ordering, which enables direct sample-wise angular distillation. We first normalize each row into a probability distribution with a row-wise softmax:

$$P_F^{a,b} = \mathrm{softmax}_b\left(\frac{M_F^{a,b}}{\tau}\right), \; P_P^{a,b} = \mathrm{softmax}_b\left(\frac{M_P^{a,b}}{\tau}\right), \tag{4}$$

where τ is a temperature parameter. To mitigate class-imbalance (head classes contributing more rows than tail classes), we adopt a class-balanced aggregation. Let n_c be the number of samples of class c in the current mini-batch, and $\mathcal{C}_B = \{\, c \in \mathcal{C}_t \mid n_c > 0 \,\}$ the set of present classes. The GSA loss is computed by averaging the row-wise KL divergence *per class* and then averaging over classes:

$$\mathcal{L}_{\mathrm{GSA}} = \frac{1}{|\mathcal{C}_B|} \sum_{c \in \mathcal{C}_B} \frac{1}{n_c} \sum_{a:\, y_a = c} \mathrm{KL}\big(P_F^{a,:} \,\|\, P_P^{a,:}\big). \tag{5}$$

This class-balanced design strengthens the alignment for tail classes during local client optimization, ensuring they receive sufficient geometric supervision. In addition, we employ a standard classification loss $\mathcal{L}_{CLS}$ to ensure the features remain discriminative and well-separated, i.e.,

$$\mathcal{L}_{\mathrm{CLS}} = -\sum_{i=1}^{C_t} y_i \log\left(e^{z_i} / \sum_{j=1}^{C_t} e^{z_j}\right) \tag{6}$$

where y_i denotes the one-hot encoded ground-truth label for class i, and $z_i = \langle \mathbf{f}, \mathbf{w}_i \rangle$ is the similarity score between the feature $\mathbf{f}$ and the prototype $\mathbf{w}_i$ of class i.

Algorithm 1 FEAT

Initialize: R: number of communication rounds; K: number of clients; T: number of tasks; $\mathcal{D}_k$: local dataset on client k, including both current and previous tasks; θ_g: global model parameters.
for each task $t = 1$ to T **do**
  for each communication round $r = 1$ to R **do**
    for each client $k = 1$ to K **do**
      Initialize local model parameters: $\theta_k \leftarrow \theta_g$
      Select a mini-batch ζ from $\mathcal{D}_k$ and update θ_k by Eq. 6 if t=1, otherwise by Eq. 15.
      Compute the local statistics $\{\bar{e}_H^{(T,k)}, \bar{e}_T^{(T,k)}\}$ via Eq. 9.
    end for
    Aggregate local models via $\theta_g = \frac{1}{N_C} \sum_k \theta_{(k)}$.
    Compute the global statistics $\{\bar{e}_H^G, \bar{e}_T^G\}$ by Eq. 10.
  end for
  Replay data in the manner of Re-Fed+ or FedCBDR.
end for
Inference: Employ the EGC module to remove bias via Eqs. 11, 12 and 13.

4.2. Energy-based Geometric Correction (EGC)

Although GSA mitigates cross-client feature misalignment, limited replay leaves a long-tailed distribution that induces a systematic drift of tail features toward head directions. Moreover, we measure this drift with *rank-normalized subspace energies* e_H and e_T (Eq. 11) computed via the projectors in Eq. 8. Obviously, a large fraction of tail samples continue to have $e_H > e_T$, signaling head-subspace bias (Figure 4). To address this, EGC performs a lightweight inference-time correction.

Task-wise ETF subspace partitioning. At incremental task t, we maintain the ETF prototypes $W_t = [W_T, W_H] \in \mathbb{R}^{d \times C_t}$ for the currently observed class set $\mathcal{C}_t$, where $W_H = [\mathbf{w}_c]_{c \in \mathcal{C}_H}$ and $W_T = [\mathbf{w}_c]_{c \in \mathcal{C}_T}$ correspond to head (current) and tail (previous) classes, respectively. Following the simplex-ETF property,

$$\mathrm{rank}(W_H) = |\mathcal{C}_H| - 1, \qquad \mathrm{rank}(W_T) = |\mathcal{C}_T| - 1. \tag{7}$$

We compute orthogonal projection operators using the Moore–Penrose pseudoinverse (†):

$$P_H = W_H (W_H^\top W_H)^\dagger W_H^\top, \qquad P_T = W_T (W_T^\top W_T)^\dagger W_T^\top. \tag{8}$$

Training-time tail priors. During local client training with exemplar replay, we estimate the typical projection energies of tail samples over the two subspaces. For a normalized feature $\tilde{f}_a$ from replayed tail data, we maintain Exponential

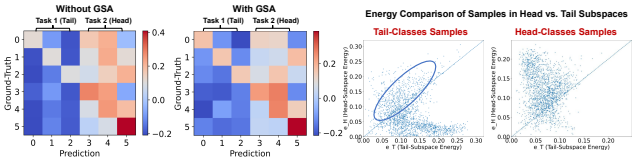

Figure 4. The GSA module reduces the tendency of tail classes to drift toward head-class directions, but tail-class features still show high similarity to head-class prototypes. Moreover, a notable fraction of tail-class samples are aligned with the head-class subspace.

Moving Averages (EMA) of the rank-normalized energies:

$$\bar{e}_H^{(T)} \leftarrow (1-\rho)\,\bar{e}_H^{(T)} + \rho \frac{1}{n_T} \sum_{a \in \mathcal{D}_T} \frac{\|P_H \tilde{f}_a\|_2^2}{|\mathcal{C}_H| - 1},$$
$$\bar{e}_T^{(T)} \leftarrow (1-\rho)\,\bar{e}_T^{(T)} + \rho \frac{1}{n_T} \sum_{a \in \mathcal{D}_T} \frac{\|P_T \tilde{f}_a\|_2^2}{|\mathcal{C}_T| - 1}. \tag{9}$$

After each local training round, clients upload only the scalar EMA statistics $\{\bar{e}_H^{(T,k)}, \bar{e}_T^{(T,k)}\}$ to the server. The server performs a sample-size–weighted aggregation:

$$\bar{e}_H^{(G)} = \frac{1}{\sum_k n_T^{(k)}} \sum_k n_T^{(k)} \bar{e}_H^{(T,k)}, \qquad \bar{e}_T^{(G)} = \frac{1}{\sum_k n_T^{(k)}} \sum_k n_T^{(k)} \bar{e}_T^{(T,k)}, \tag{10}$$

where k indexes clients and $n_T^{(k)}$ is the number of replayed tail samples used for the local EMA update.

Inference-time decontamination. During inference, for any normalized feature $\tilde{x}$, we compute rank-normalized projection energies:

$$e_H(\tilde{x}) = \frac{\|P_H \tilde{x}\|_2^2}{|\mathcal{C}_H| - 1}, \qquad e_T(\tilde{x}) = \frac{\|P_T \tilde{x}\|_2^2}{|\mathcal{C}_T| - 1}. \tag{11}$$

A confidence gate is derived from the deviation above the global tail prior:

$$g(\tilde{x}) = \max\left\{ \frac{e_H(\tilde{x}) - \bar{e}_H^{(G)}}{e_H(\tilde{x}) + e_T(\tilde{x}) + \varepsilon},\ 0 \right\}. \tag{12}$$

The corrected representation suppresses the head-aligned component and enhances the tail-aligned counterpart, followed by ℓ_2-normalization:

$$\tilde{x}' = \tilde{x} - g(\tilde{x}) P_H \tilde{x} + g(\tilde{x}) P_T \tilde{x}, \qquad \tilde{x}' \leftarrow \frac{\tilde{x}'}{\|\tilde{x}'\|_2}. \tag{13}$$

Finally, prediction is made with ETF-based similarities:

$$z_c = (\tilde{x}')^\top \mathbf{w}_c, \qquad c \in \mathcal{C}_t. \tag{14}$$

4.3. Training Strategy

FEAT adopts a stage-dependent optimization scheme:

- **Initial task** ($t = 1$)**:** Only the classification loss is applied: $\mathcal{L} = \mathcal{L}_{\text{CLS}}$.
- **Subsequent tasks** ($t > 1$)**:** Both the classification loss and the GSA loss are employed:

$$\mathcal{L} = \mathcal{L}_{\text{CLS}} + \lambda \cdot \mathcal{L}_{\text{GSA}}, \tag{15}$$

where λ is a balancing hyperparameter.

5. Experiments

5.1. Experiment Settings

Datasets. Following prior work [19, 50], we use three widely used benchmarks: CIFAR10/100 [15], and TinyImageNet-Subset [16]. To emulate non-iid client environments, we partition local datasets using a Dirichlet distribution. Specifically, we consider two client settings (5 and 10 clients) for all datasets, and two task granularities per dataset: CIFAR10 is split into 3 and 5 tasks with $\beta = \{0.5, 1.0\}$, while CIFAR100 and TinyImageNet-Subset are split into 5 and 10 tasks with $\beta = \{0.1, 0.5\}$.

Evaluation Metric. Following prior work [19, 25], we report Top-1 Accuracy, defined as $N_{\text{correct}}/N_{\text{total}}$, where N_{correct} and N_{total} denote the numbers of correct predictions and samples, respectively.

Implementation Details. We use ResNet-18 as the backbone for all datasets, and the classifier expands as new classes arrive. Each client trains for 2 local epochs per communication round with batch size 128. We run 100 communication rounds per task using SGD with learning rate 0.04 and weight decay 1×10^{-5}. The exemplar budget per task is 450/300 for CIFAR10 (3/5 tasks), 1000/500 for CIFAR100 (5/10 tasks), and 500/250 for TinyImageNet-Subset (5/10 tasks). For GSA, the temperature τ is chosen from $\{0.07, 0.5\}$ and the weight λ from $\{0.05, 0.1, 0.5\}$. For EGC, the EMA decay ρ is selected from $\{0.5, 0.7, 0.9\}$. We set the numerical stability term ε to 10^{-8}.

5.2. Performance Comparison

To validate the effectiveness of the method FEAT, we compare it with seven state-of-the-art baselines, including FedEWC [14], FedLwF [22], TARGET [50], LANDER [43], FedSSI [21], Re-Fed+ [20], and FedCBDR [36]. To explore the adaptability of FEAT in different frameworks, we construct two hybrid versions: FEAT_R and FEAT_F, which correspond to incorporating FEAT into Re-Fed+ and FedCBDR, respectively. Table 1 summarizes the outcomes:

- FEAT_F consistently achieves the highest Top-1 accuracy across all cases, including various levels of heterogeneity, and different task partitions, demonstrating the robustness of the proposed de-biased framework.
- Both FEAT_R and FEAT_F achieve performance improvements over their corresponding baselines, which validates the importance of the core idea behind our method, which also highlights the plug-and-play nature.
- As the setting becomes more challenging, with stronger heterogeneity and more clients, FEAT's advantage over other methods does not shrink. This indicates that its gains are not tied to easy cases but remain stable when scaled to more complex and realistic FCL scenarios.

Table 1. Performance comparison between FEAT and baselines across three datasets under varying levels of data heterogeneity (β) and different numbers of clients (K). Results are averaged over three random seeds, with standard deviations reported.

Method	CIFAR10				CIFAR100				TinyImageNet-Subset			
	3 Tasks		5 Tasks		5 Tasks		10 Tasks		5 Tasks		10 Tasks	
	β=0.5	β=1.0	β=0.5	β=1.0	β=0.1	β=0.5	β=0.1	β=0.5	β=0.1	β=0.5	β=0.1	β=0.5
	$K = 5$ **Clients**											
FedEWC	$40.48_{\pm1.6}$	$43.11_{\pm3.1}$	$22.16_{\pm2.1}$	$29.38_{\pm2.3}$	$18.66_{\pm2.6}$	$22.64_{\pm4.4}$	$10.12_{\pm0.6}$	$11.34_{\pm0.6}$	$11.52_{\pm1.4}$	$14.27_{\pm2.2}$	$7.92_{\pm0.7}$	$8.85_{\pm0.9}$
FedLwF	$56.84_{\pm1.2}$	$58.97_{\pm3.1}$	$39.21_{\pm1.9}$	$52.32_{\pm2.7}$	$34.45_{\pm2.0}$	$36.34_{\pm2.6}$	$19.36_{\pm1.4}$	$24.91_{\pm0.9}$	$16.78_{\pm0.8}$	$21.87_{\pm0.7}$	$12.16_{\pm0.6}$	$13.11_{\pm0.3}$
TARGET	$43.53_{\pm3.8}$	$54.87_{\pm4.2}$	$36.02_{\pm2.4}$	$48.91_{\pm1.6}$	$30.62_{\pm2.9}$	$32.89_{\pm3.7}$	$13.04_{\pm1.1}$	$20.57_{\pm0.7}$	$17.69_{\pm1.5}$	$21.11_{\pm1.3}$	$11.98_{\pm0.5}$	$15.28_{\pm0.7}$
LANDER	$54.41_{\pm2.7}$	$59.88_{\pm1.9}$	$39.63_{\pm2.0}$	$57.55_{\pm2.7}$	$43.59_{\pm2.9}$	$48.39_{\pm3.3}$	$27.36_{\pm1.1}$	$32.64_{\pm2.8}$	$11.91_{\pm1.2}$	$24.77_{\pm2.5}$	$12.82_{\pm0.9}$	$14.93_{\pm1.2}$
Re-Fed+	$52.95_{\pm4.1}$	$61.64_{\pm3.9}$	$54.15_{\pm2.7}$	$58.94_{\pm3.2}$	$31.92_{\pm4.3}$	$37.68_{\pm2.3}$	$28.64_{\pm0.8}$	$38.62_{\pm1.7}$	$22.89_{\pm0.4}$	$26.07_{\pm0.5}$	$16.71_{\pm1.3}$	$21.02_{\pm1.4}$
FedSSI	$51.44_{\pm3.1}$	$56.33_{\pm3.1}$	$52.26_{\pm2.5}$	$56.49_{\pm2.5}$	$31.27_{\pm1.3}$	$37.63_{\pm1.3}$	$26.71_{\pm1.3}$	$31.16_{\pm1.3}$	$18.56_{\pm1.1}$	$21.37_{\pm1.1}$	$14.41_{\pm0.9}$	$17.22_{\pm0.9}$
FedCBDR	$63.32_{\pm1.6}$	$65.88_{\pm2.1}$	$61.77_{\pm1.5}$	$64.79_{\pm1.1}$	$45.84_{\pm2.0}$	$50.33_{\pm2.3}$	$44.52_{\pm1.7}$	$45.96_{\pm1.3}$	$25.22_{\pm1.1}$	$26.38_{\pm1.6}$	$19.03_{\pm0.7}$	$21.98_{\pm0.5}$
$\texttt{FEAT}_R$	$60.18_{\pm1.4}$	$70.28_{\pm1.6}$	$60.38_{\pm1.5}$	$64.62_{\pm1.6}$	$37.14_{\pm1.8}$	$42.31_{\pm1.4}$	$32.91_{\pm1.3}$	$43.34_{\pm1.9}$	$23.31_{\pm1.5}$	$27.36_{\pm1.6}$	$19.26_{\pm1.7}$	$22.74_{\pm1.1}$
$\texttt{FEAT}_F$	$\mathbf{72.67}_{\pm1.5}$	$\mathbf{74.21}_{\pm2.6}$	$\mathbf{70.19}_{\pm1.8}$	$\mathbf{73.63}_{\pm2.0}$	$\mathbf{50.14}_{\pm1.2}$	$\mathbf{53.31}_{\pm0.9}$	$\mathbf{48.16}_{\pm1.6}$	$\mathbf{49.18}_{\pm0.8}$	$\mathbf{25.88}_{\pm1.6}$	$\mathbf{29.31}_{\pm0.6}$	$\mathbf{19.41}_{\pm1.4}$	$\mathbf{23.45}_{\pm0.4}$
	$K = 10$ **Clients**											
FedEWC	$35.11_{\pm3.5}$	$43.32_{\pm2.3}$	$22.13_{\pm2.0}$	$24.97_{\pm2.8}$	$16.95_{\pm0.4}$	$17.21_{\pm0.4}$	$5.41_{\pm1.2}$	$12.28_{\pm0.6}$	$10.73_{\pm1.3}$	$14.37_{\pm2.4}$	$5.42_{\pm1.1}$	$6.47_{\pm0.6}$
FedLwF	$49.72_{\pm2.0}$	$47.51_{\pm3.1}$	$41.63_{\pm2.4}$	$48.35_{\pm3.9}$	$28.76_{\pm1.7}$	$36.35_{\pm1.9}$	$12.55_{\pm1.3}$	$18.91_{\pm0.8}$	$17.34_{\pm0.6}$	$21.79_{\pm1.7}$	$5.28_{\pm1.1}$	$11.04_{\pm1.0}$
TARGET	$39.20_{\pm2.6}$	$40.18_{\pm1.2}$	$22.45_{\pm1.9}$	$27.79_{\pm1.5}$	$17.89_{\pm1.4}$	$22.67_{\pm1.7}$	$12.72_{\pm1.2}$	$15.98_{\pm1.1}$	$19.12_{\pm1.6}$	$25.88_{\pm0.5}$	$6.23_{\pm1.4}$	$10.74_{\pm1.0}$
LANDER	$42.38_{\pm3.0}$	$44.89_{\pm1.9}$	$26.63_{\pm4.2}$	$31.47_{\pm2.8}$	$29.92_{\pm2.5}$	$42.83_{\pm2.7}$	$11.60_{\pm1.4}$	$26.49_{\pm1.5}$	$15.68_{\pm2.7}$	$21.90_{\pm2.1}$	$10.38_{\pm1.4}$	$10.68_{\pm1.2}$
Re-Fed+	$44.92_{\pm2.2}$	$52.97_{\pm3.9}$	$37.84_{\pm1.8}$	$40.16_{\pm1.6}$	$32.77_{\pm1.9}$	$35.84_{\pm0.4}$	$28.67_{\pm2.2}$	$32.74_{\pm0.6}$	$20.26_{\pm1.7}$	$25.67_{\pm1.8}$	$15.46_{\pm1.3}$	$19.32_{\pm1.6}$
FedSSI	$41.93_{\pm3.1}$	$48.27_{\pm3.1}$	$38.22_{\pm2.5}$	$40.45_{\pm2.5}$	$30.56_{\pm1.3}$	$35.72_{\pm1.3}$	$27.54_{\pm1.3}$	$33.16_{\pm1.3}$	$16.55_{\pm1.1}$	$20.27_{\pm1.1}$	$11.32_{\pm0.9}$	$14.49_{\pm0.9}$
FedCBDR	$59.27_{\pm1.9}$	$61.75_{\pm1.1}$	$52.00_{\pm2.5}$	$59.91_{\pm1.0}$	$41.96_{\pm1.5}$	$47.54_{\pm1.9}$	$37.11_{\pm1.0}$	$43.59_{\pm2.4}$	$22.95_{\pm0.5}$	$25.45_{\pm1.9}$	$17.61_{\pm1.1}$	$19.54_{\pm1.0}$
$\texttt{FEAT}_R$	$54.62_{\pm1.5}$	$62.34_{\pm2.1}$	$46.59_{\pm1.3}$	$49.53_{\pm1.1}$	$35.75_{\pm1.8}$	$40.43_{\pm1.6}$	$32.38_{\pm1.1}$	$37.79_{\pm1.4}$	$23.11_{\pm1.9}$	$27.15_{\pm1.1}$	$18.16_{\pm1.0}$	$20.79_{\pm0.9}$
$\texttt{FEAT}_F$	$\mathbf{70.63}_{\pm1.7}$	$\mathbf{72.89}_{\pm1.9}$	$\mathbf{65.48}_{\pm2.1}$	$\mathbf{69.76}_{\pm1.4}$	$\mathbf{45.72}_{\pm1.6}$	$\mathbf{50.36}_{\pm1.4}$	$\mathbf{39.36}_{\pm1.8}$	$\mathbf{47.81}_{\pm2.1}$	$\mathbf{25.15}_{\pm0.9}$	$\mathbf{28.85}_{\pm1.0}$	$\mathbf{20.49}_{\pm1.6}$	$\mathbf{23.17}_{\pm1.1}$

Table 2. Ablation studies are conducted on CIFAR-10 with 3 tasks under $\beta \in \{0.5, 1.0\}$ and on CIFAR-100 with 5 tasks under $\beta \in \{0.1, 0.5\}$. The number of clients is 5.

Method	CIFAR10		CIFAR100	
	β=0.5	β=1.0	β=0.1	β=0.5
FedCBDR	$63.32_{\pm1.6}$	$65.88_{\pm2.1}$	$45.84_{\pm2.0}$	$50.33_{\pm2.3}$
+ETF	$62.17_{\pm1.6}$	$63.66_{\pm2.0}$	$44.58_{\pm1.9}$	$48.69_{\pm1.2}$
+ETF+GSA	$68.54_{\pm1.6}$	$70.42_{\pm1.7}$	$47.77_{\pm1.2}$	$52.23_{\pm1.6}$
+ETF+EGC	$69.12_{\pm1.8}$	$70.83_{\pm2.1}$	$47.16_{\pm1.4}$	$51.72_{\pm1.5}$
+ETF+GSA+EGC	$\mathbf{72.67}_{\pm1.5}$	$\mathbf{74.21}_{\pm2.6}$	$\mathbf{50.14}_{\pm1.2}$	$\mathbf{53.31}_{\pm0.9}$

- Although generative replay method LANDER achieves competitive performance compared to exemplar replay-based methods in some cases, they rely heavily on a large number of generated samples, which poses potential limitations when deployed in large-scale environments.

5.3. Ablation Study

This section aims to investigate the contributions of key components, including the Geometric Structure Alignment (GSA) module and the Energy-based Geometric Correction (EGC) module. The results are presented in Table 3:

- Relying solely on the ETF classifier may not yield performance gains, as severe data imbalance substantially undermines cross-source representation alignment.
- The integration of GSA consistently enhances performance under all datasets and task settings. This improvement stems from GSA's capacity to maintain inter-class geometric consistency, which helps reduce the representation gap among heterogeneous clients.

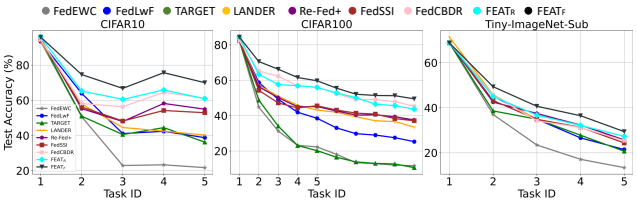

Figure 5. Evaluation on CIFAR-10 (5 tasks), CIFAR-100 (10 tasks), and TinyImageNet-Subset (5 tasks) with 5 clients and heterogeneity $\beta = 0.5$. $\texttt{FEAT}_F$ and $\texttt{FEAT}_R$ consistently outperform their baselines on all cases.

- By addressing task-level imbalance during inference, the EGC module stabilizes predictions, improves performance, and normalizes decision boundaries, reducing head-task dominance.

5.4. Forgetting Curve across Incremental Tasks

This section evaluates FEAT and baselines on three datasets with heterogeneity $\beta = 0.5$. As shown in Figure 5, $\texttt{FEAT}_F$ and $\texttt{FEAT}_R$ consistently outperform their baselines across all increments by mitigating client heterogeneity and task-level imbalance. Moreover, $\texttt{FEAT}_F$ attains the highest initial accuracy and shows the slowest, most stable decline, indicating stronger long-term resistance to forgetting.

5.5. Communication Cost Analysis

FEAT retains the same training rounds and aggregation protocol as the baseline, adding only negligible overhead. Per

round, each client uploads two scalar EMA statistics ($\bar{e}_H$ and $\bar{e}_T$), which is trivial compared to model parameters, and the server performs weighted aggregation. GSA runs locally and requires no extra transmission, while EGC is applied at inference and introduces no training-time communication. Overall, the per-round cost is approximately the baseline model upload plus $O(1)$ scalars.

5.6. Analyzing the Influence of Replay Data Volume

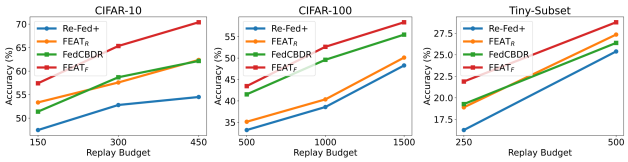

Figure 6. Impact of per-task memory capacity M on model performance across different datasets, where $\texttt{FEAT}_F$ shows the best overall performance.

This section evaluates $\texttt{FEAT}_F$ and $\texttt{FEAT}_R$ against baselines under different replay budgets on three datasets (5-task split), with replay sizes set to $\{100, 150, 300\}$ (CIFAR-10), $\{125, 250, 500\}$ (CIFAR-100), and $\{250, 500\}$ (TinyImageNet-Subset). As shown in Figure 6, $\texttt{FEAT}_F$ and $\texttt{FEAT}_R$ consistently achieve performance improvements over their respective baselines across all settings. These results confirm the effectiveness of our design and emphasize the importance of tackling client heterogeneity and task-level imbalance in FCL. Compared to $\texttt{FEAT}_R$, $\texttt{FEAT}_F$ shows more significant performance improvements as the replay budget increases. This can be attributed to the class-wise balanced replay inherited from FedCBDR.

5.7. Sensitivity Study of Hyper-parameters

We assess hyperparameter sensitivity on CIFAR-10 (5 tasks, β=0.5, 5 clients), varying $\lambda \in \{1, 5, 10\}$, $\rho \in \{0.5, 0.7, 0.9\}$, and $\tau \in \{0.1, 0.5\}$. Figure 7 shows stable accuracy with only slight fluctuations, indicating robustness to hyperparameter choices. In general, a smaller λ better exploits geometric structure knowledge, a larger ρ yields more stable debiased learning, and τ has a minor effect. Notably, these configurations consistently outperform its baseline.

5.8. Performance Analysis of Alternative Strategies

This section evaluates several commonly used enhancements on top of the same baseline (FedCBDR), including MOON [17], FedRCL [39], CLIP2FL [41], FedFSA [35], and FEAT. Table 3 reports the mean accuracy over three runs on three datasets under a 5-task setting with 5 clients. Across all cases, FEAT achieves the highest accuracy when added to FedCBDR. The gains are especially clear in the more heterogeneous cases (smaller β), where

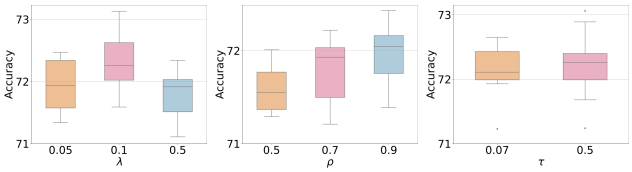

Figure 7. Sensitivity analysis of hyperparameters λ, ρ, and τ, adjusted over the ranges $\{0.05, 0.1, 0.5\}$, $\{0.5, 0.7, 0.9\}$, and $\{0.07, 0.5\}$, respectively. For each parameter, the default value is fixed while the other two are varied. It achieves robust performance under a wide range of these parameters.

Table 3. Performance analysis of alternative optimization strategies, including MOON, FedRCL, CLIP2FL, FedFSA, and FEAT, applied on top of the FedCBDR baseline. Results are reported as the mean over three runs on three datasets.

Methods	CIFAR10		CIFAR100		Tiny-Subset	
	$\beta = 0.5$	$\beta = 1.0$	$\beta = 0.1$	$\beta = 0.5$	$\beta = 0.1$	$\beta = 0.5$
FedCBDR	61.77	64.79	45.84	50.33	25.22	26.38
+ MOON	62.56	66.32	45.14	49.32	24.73	26.44
+ FedRCL	58.74	60.21	42.19	46.46	20.27	22.46
+ CLIP2FL	62.33	67.97	46.19	51.11	24.18	26.67
+ FedFSA	64.42	68.32	46.88	51.38	24.61	26.33
+ FEAT	**70.19**	**73.63**	**50.14**	**53.31**	**25.88**	**29.31**

other strategies provide only moderate or inconsistent improvements, while FEAT delivers a consistent boost. This suggests that FEAT is effective at handling spatiotemporal drift and cross-client imbalance, rather than only refining local features. In contrast, methods such as MOON and FedFSA offer incremental benefits but do not close the gap.

5.9. Final Performance Comparison Across Tasks

Table 4. Per-task and average accuracy (%) of different methods on CIFAR10 and CIFAR100.

	CIFAR10			CIFAR100				
	Task 1	Task 2	Task 3	Task 1	Task 2	Task 3	Task 4	Task 5
FedCBDR	43.37	33.90	96.80	43.80	34.25	32.85	45.00	83.55
+ GSA	47.47	51.60	**97.92**	48.75	36.25	36.10	52.15	**84.40**
+ EGC	53.53	53.57	97.65	50.15	38.70	39.10	51.40	82.45
+ EGC + GSA	**60.83**	**69.97**	94.28	**50.70**	**40.65**	**39.70**	**53.85**	81.95

This section evaluates the final model on each task for CIFAR-10 (3 tasks) and CIFAR-100 (5 tasks) under non-IID heterogeneity β=0.5 with 5 clients. As shown in Table 4, adding GSA and EGC markedly improves final performance on earlier tasks, with the best results when combined. Although accuracy on the last task slightly decreases, it remains competitive, indicating stronger resistance to forgetting and more stable task-wise generalization.

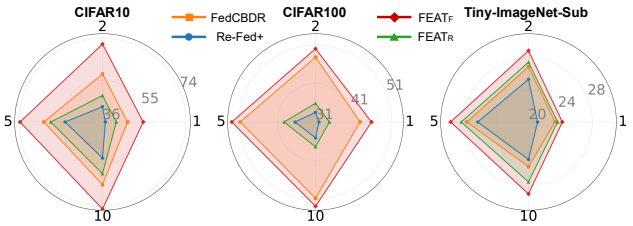

Figure 8. Performance of Re-Fed+ and FedCBDR, with and without FEAT, across local epochs $\{1, 2, 5, 10\}$ on three datasets. And FEAT brings performance gains to all baselines.

5.10. Evaluation under Different Local Epochs

This section evaluates the performance of Re-Fed+ and FedCBDR, with and without FEAT, under varying local epochs $\{1, 2, 5, 10\}$. As shown in Figure 8, the versions integrated with FEAT consistently outperform their baselines in all settings. However, the performance improvement does not exhibit a strictly increasing trend with more local updates, indicating that the advantages of GSA and EGC may saturate or become unstable when local training dominates the optimization process. This observation provides valuable insight for future work.

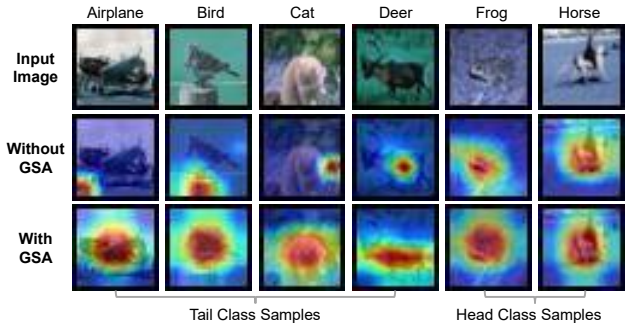

Figure 9. Visualization of visual attention. The GSA module corrects the attention bias of tail classes and strengthens the confidence of attention on head classes.

5.11. Comparative Analysis of Visual Attention

Figure 4 shows that GSA reduces tail-to-head drift. This section provides complementary qualitative evidence by visualizing attention maps for tail and head samples. As shown in Figure 9, without GSA, tail classes show diffuse or off-target responses, indicating that the model is not attending reliably to the true object. With GSA, attention becomes tighter and aligned with the object region, correcting this bias. For head classes, GSA further sharpens and amplifies the focus around the object, suggesting increased confidence rather than degradation. These results demonstrate the positive impact of the GSA module.

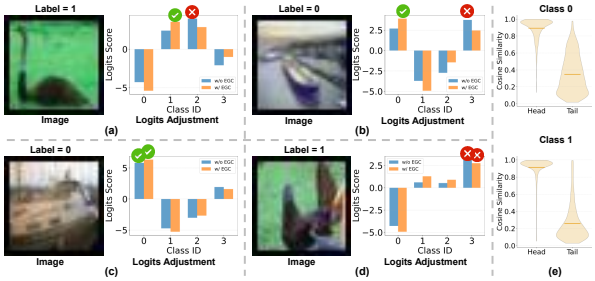

Figure 10. Geometric correction effect of the EGC module on the CIFAR-10 (first two tasks). (a–b) EGC corrects the prediction bias; (c) EGC corrects the prediction bias; (d) it struggles when features deviate too far from the ground truth; (e) The removed component aligns more with the head subspace than the tail subspace for both Class 0 and Class 1.

5.12. Logits-Level Behavior Analysis

This section further analyzes how the EGC module adjusts logits at inference time. Figure 10 shows four representative CIFAR-10 samples from the first two tasks. Obviously, cases (a) and (b) show that applying EGC amplifies the logit of the target class and suppresses the spurious head-class logits, leading to a correct prediction. Case (c) shows that EGC further enlarges the gap between the ground-truth class and the other classes. Case (d) illustrates a failure case, where the feature has drifted far from the true class manifold; in this situation, the correction is not sufficient to fully recover the correct label, although it still narrows the gap between the ground-truth class and the current top-1 class. In case (e), for tail-class samples, the removed component aligns strongly with the head-class subspace and weakly with the tail subspace, confirming that our correction targets head-attraction drift rather than a uniform shift.

6. Conclusions and Future Work

To address the dual challenges of inter-client heterogeneity and task-level data imbalance, we propose FEAT, which performs bias mitigation through both prototype-level alignment and calibrated classification. Specifically, it consists of two key components: (1) the GSA module that enforces structural consistency between local features and globally shared prototypes, thus promoting inter-client alignment; and (2) the EGC that removes projection bias during inference, improving robustness under class-imbalanced data distributions. Experimental results show that FEAT consistently outperforms existing methods by effectively mitigating inter-client and inter-task bias.

In the future, we plan to explore more advanced cross-client representation alignment techniques that can better adapt to dynamically evolving heterogeneity. Additionally, we intend to improve the bias mitigation capability of the classifier by integrating uncertainty estimation techniques.

References

[1] Sara Babakniya, Zalan Fabian, Chaoyang He, Mahdi Soltanolkotabi, and Salman Avestimehr. A data-free approach to mitigate catastrophic forgetting in federated class incremental learning for vision tasks. *Advances in Neural Information Processing Systems*, 36:66408–66425, 2023. 1

[2] Sara Babakniya, Zalan Fabian, Chaoyang He, Mahdi Soltanolkotabi, and Salman Avestimehr. Don't memorize; mimic the past: Federated class incremental learning without episodic memory. *arXiv preprint arXiv:2307.00497*, 2023. 2

[3] Syreen Banabilah, Moayad Aloqaily, Eitaa Alsayed, Nida Malik, and Yaser Jararweh. Federated learning review: Fundamentals, enabling technologies, and future applications. *Information processing & management*, 59(6):103061, 2022. 1

[4] Tanujit Chakraborty, Ujjwal Reddy KS, Shraddha M Naik, Madhurima Panja, and Bayapureddy Manvitha. Ten years of generative adversarial nets (gans): a survey of the state-of-the-art. *Machine Learning: Science and Technology*, 5(1): 011001, 2024. 1

[5] Yuanlu Chen, Alysa Ziying Tan, Siwei Feng, Han Yu, Tao Deng, Libang Zhao, and Feng Wu. General federated class-incremental learning with lightweight generative replay. *IEEE Internet of Things Journal*, 2024. 2

[6] Jiahua Dong, Lixu Wang, Zhen Fang, Gan Sun, Shichao Xu, Xiao Wang, and Qi Zhu. Federated class-incremental learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10164–10173, 2022. 2

[7] Ziqing Fan, Jiangchao Yao, Bo Han, Ya Zhang, Yanfeng Wang, et al. Federated learning with bilateral curation for partially class-disjoint data. *Advances in Neural Information Processing Systems*, 36:32006–32019, 2023. 3

[8] Lele Fu, Sheng Huang, Yuecheng Li, Chuan Chen, Chuanfu Zhang, and Zibin Zheng. Learn the global prompt in the low-rank tensor space for heterogeneous federated learning. *Neural Networks*, 187:107319, 2025. 1

[9] Laurent Girin, Simon Leglaive, Xiaoyu Bie, Julien Diard, Thomas Hueber, and Xavier Alameda-Pineda. Dynamical variational autoencoders: A comprehensive review. *arXiv preprint arXiv:2008.12595*, 2020. 2

[10] Ian J Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. *Advances in neural information processing systems*, 27, 2014. 2

[11] Ming Hu, Zhihao Yue, Xiaofei Xie, Cheng Chen, Yihao Huang, Xian Wei, Xiang Lian, Yang Liu, and Mingsong Chen. Is aggregation the only choice? federated learning via layer-wise model recombination. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 1096–1107, 2024. 1

[12] Ming Hu, Peiheng Zhou, Zhihao Yue, Zhiwei Ling, Yihao Huang, Anran Li, Yang Liu, Xiang Lian, and Mingsong Chen. Fedcross: Towards accurate federated learning via multi-model cross-aggregation. In *IEEE International Conference on Data Engineering (ICDE)*, pages 2137–2150. IEEE, 2024. 1

[13] Peter Kairouz, H Brendan McMahan, Brendan Avent, Aurélien Bellet, Mehdi Bennis, Arjun Nitin Bhagoji, Kallista Bonawitz, Zachary Charles, Graham Cormode, Rachel Cummings, et al. Advances and open problems in federated learning. *Foundations and trends® in machine learning*, 14(1–2): 1–210, 2021. 1

[14] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the national academy of sciences*, 114(13):3521–3526, 2017. 5

[15] Alex Krizhevsky, Geoffrey Hinton, et al. Learning multiple layers of features from tiny images. 2009. 5

[16] Ya Le and Xuan Yang. Tiny imagenet visual recognition challenge. *CS 231N*, 7(7):3, 2015. 5

[17] Qinbin Li, Bingsheng He, and Dawn Song. Model-contrastive federated learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10713–10722, 2021. 7

[18] Qi Li, Yucan Zhou, Jiang Zhou, XiaoYan Gu, and Bo Li. Tackling feature skew in heterogeneous federated learning with semantic enhancement. In *2024 IEEE International Conference on Multimedia and Expo (ICME)*, pages 1–6. IEEE, 2024. 3

[19] Yichen Li, Qunwei Li, Haozhao Wang, Ruixuan Li, Wenliang Zhong, and Guannan Zhang. Towards efficient replay in federated incremental learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12820–12829, 2024. 2, 5

[20] Yichen Li, Haozhao Wang, Yining Qi, Wei Liu, and Ruixuan Li. Re-fed+: A better replay strategy for federated incremental learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025. 2, 3, 5

[21] Yichen Li, Yuying Wang, Haozhao Wang, Yining Qi, Tianzhe Xiao, and Ruixuan Li. Rehearsal-free continual federated learning with synergistic synaptic intelligence. *International Conference on Machine Learning*, 2025. 5

[22] Zhizhong Li and Derek Hoiem. Learning without forgetting. *IEEE transactions on pattern analysis and machine intelligence*, 40(12):2935–2947, 2017. 5

[23] Zexi Li, Xinyi Shang, Rui He, Tao Lin, and Chao Wu. No fear of classifier biases: Neural collapse inspired federated learning with synthetic and fixed classifier. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5319–5329, 2023. 3

[24] Jinglin Liang, Jin Zhong, Hanlin Gu, Zhongqi Lu, Xingxing Tang, Gang Dai, Shuangping Huang, Lixin Fan, and Qiang Yang. Diffusion-driven data replay: A novel approach to combat forgetting in federated class continual learning. In *European Conference on Computer Vision*, pages 303–319. Springer, 2024. 2

[25] Tianchi Liao, Lele Fu, Jialong Chen, Zhen Wang, Zibin Zheng, and Chuan Chen. A swiss army knife for heterogeneous federated learning: Flexible coupling via trace norm. *Advances in Neural Information Processing Systems*, 37: 139886–139911, 2024. 5

[26] Jianfeng Lu and Stefan Steinerberger. Neural collapse under cross-entropy loss. *Applied and Computational Harmonic Analysis*, 59:224–241, 2022. 3

[27] Yanbiao Ma, Wei Dai, Wenke Huang, and Jiayi Chen. Geometric knowledge-guided localized global distribution alignment for federated learning. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 20958–20968, 2025. 1

[28] Hojjat Navidan, Parisa Fard Moshiri, Mohammad Nabati, Reza Shahbazian, Seyed Ali Ghorashi, Vahid Shah-Mansouri, and David Windridge. Generative adversarial networks (gans) in networking: A comprehensive survey & evaluation. *Computer Networks*, 194:108149, 2021. 1

[29] Thinh Nguyen, Khoa D Doan, Binh T Nguyen, Danh Le-Phuoc, and Kok-Seng Wong. Overcoming catastrophic forgetting in federated class-incremental learning via federated global twin generator. *arXiv preprint arXiv:2407.11078*, 2024. 2

[30] Milad Khademi Nori, Il-Min Kim, and Guanghui Wang. Federated class-incremental learning: A hybrid approach using latent exemplars and data-free techniques to address local and global forgetting. *arXiv preprint arXiv:2501.15356*, 2025. 2

[31] Vardan Papyan, XY Han, and David L Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences*, 117(40):24652–24663, 2020. 3

[32] Lucas Pinheiro Cinelli, Matheus Araújo Marins, Eduardo Antúnio Barros da Silva, and Sérgio Lima Netto. Variational autoencoder. In *Variational methods for machine learning with applications to deep networks*, pages 111–149. Springer, 2021. 2

[33] Daiqing Qi, Handong Zhao, and Sheng Li. Better generative replay for continual federated learning. *arXiv preprint arXiv:2302.13001*, 2023. 1

[34] Zhuang Qi, Lei Meng, Zitan Chen, Han Hu, Hui Lin, and Xiangxu Meng. Cross-silo prototypical calibration for federated learning with non-iid data. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages 3099–3107, 2023. 1

[35] Zhuang Qi, Lei Meng, and et al. Cross-silo feature space alignment for federated learning on clients with imbalanced data. In *The 39th Annual AAAI Conference on Artificial Intelligence (AAAI-25)*, pages 19986–19994, 2025. 7

[36] Zhuang Qi, Ying-Peng Tang, Lei Meng, Han Yu, Xiaoxiao Li, and Xiangxu Meng. Class-wise balancing data replay for federated class-incremental learning. In *The Thirty-Ninth Annual Conference on Neural Information Processing Systems*, 2025. 2, 3, 5

[37] Xuankun Rong, Jianshu Zhang, Kun He, and Mang Ye. Can: Leveraging clients as navigators for generative replay in federated continual learning. ICML, 2025. 1

[38] Minhyuk Seo, Hyunseo Koh, Wonje Jeung, Minjae Lee, San Kim, Hankook Lee, Sungjun Cho, Sungik Choi, Hyunwoo Kim, and Jonghyun Choi. Learning equi-angular representations for online continual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23933–23942, 2024. 3

[39] Seonguk Seo, Jinkyu Kim, Geeho Kim, and Bohyung Han. Relaxed contrastive learning for federated learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12279–12288, 2024. 7

[40] Donald Shenaj, Marco Toldo, Alberto Rigon, and Pietro Zanuttigh. Asynchronous federated continual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5055–5063, 2023. 1

[41] Jiangming Shi, Shanshan Zheng, Xiangbo Yin, Yang Lu, Yuan Xie, and Yanyun Qu. Clip-guided federated learning on heterogeneity and long-tailed data. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 14955–14963, 2024. 7

[42] Rui Sun, Yumin Zhang, Varun Ojha, Tejal Shah, Haoran Duan, Bo Wei, and Rajiv Ranjan. Exemplar-condensed federated class-incremental learning. *arXiv preprint arXiv:2412.18926*, 2024. 2

[43] Minh-Tuan Tran, Trung Le, Xuan-May Le, Mehrtash Harandi, and Dinh Phung. Text-enhanced data-free approach for federated class-incremental learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23870–23880, 2024. 2, 5

[44] Haozhao Wang, Haoran Xu, Yichen Li, Yuan Xu, Ruixuan Li, and Tianwei Zhang. Fedcda: Federated learning with cross-rounds divergence-aware aggregation. In *The Twelfth International Conference on Learning Representations*, 2023. 1

[45] Naibo Wang, Yuchen Deng, Wenjie Feng, Jianwei Yin, and See-Kiong Ng. Data-free federated class incremental learning with diffusion-based generative memory. *arXiv preprint arXiv:2405.17457*, 2024. 1, 2

[46] Feng Wu, Alysa Ziying Tan, Siwei Feng, Han Yu, Tao Deng, Libang Zhao, and Yuanlu Chen. Federated class-incremental learning via weighted aggregation and distillation. *IEEE Internet of Things Journal*, 2025. 1

[47] Xin Yang, Hao Yu, Xin Gao, Hao Wang, Junbo Zhang, and Tianrui Li. Federated continual learning via knowledge fusion: A survey. *IEEE Transactions on Knowledge and Data Engineering*, 36(8):3832–3850, 2024. 2

[48] Hao Yu, Xin Yang, Xin Gao, Yihui Feng, Hao Wang, Yan Kang, and Tianrui Li. Overcoming spatial-temporal catastrophic forgetting for federated class-incremental learning. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pages 5280–5288, 2024. 1, 2

[49] Chengchao Zhang, Fanhua Shang, Hongying Liu, Liang Wan, and Wei Feng. Fedagc: Federated continual learning with asymmetric gradient correction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 3841–3850, 2025. 1

[50] Jie Zhang, Chen Chen, Weiming Zhuang, and Lingjuan Lyu. Target: Federated class-continual learning via exemplar-free distillation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4782–4793, 2023. 2, 5

[51] Jinghua Zhang, Li Liu, Olli Silvén, Matti Pietikäinen, and Dewen Hu. Few-shot class-incremental learning for classification and object detection: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025. 2